# Quantum Semantics: Prime Resonance, Entropy Collapse, and Conceptual Coherence

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**Abstract**

We present a framework for *quantum semantics*, extending earlier work on representing conceptual relationships through quantum field theoretic principles. Concepts are modeled as states in a prime-based Hilbert space, with resonance and coherence operators governing their interactions. Entropy functions define semantic collapse and stabilization, revealing structured quantum-like dynamics in meaning itself. Experimental simulations demonstrate entropy stabilization curves, harmonic alignment, and cross-cultural coherence patterns. Applications include semantic search, non-local semantic communication, cryptographic semantics, and consciousness-integrated artificial intelligence. This framework positions semantics not as an emergent feature of cognition, but as a fundamental resonance structure of consciousness and reality.

## 1 Introduction

Traditional semantic analysis—from vector embeddings to graph-based methods—captures useful relationships but fails to address the intrinsic dynamism and coherence of meaning. Building on the foundation of quantum cognition and resonance-based theories, we extend *quantum semantics* into a rigorous mathematical framework grounded in prime-based Hilbert spaces and entropy-driven collapse. Earlier work demonstrated the viability of representing concepts as quantum wavefunctions and analyzing resonance patterns [1]. Since then, advances in prime-based resonance theory [2, 6] and entropy-driven quantum encoding [3] provide a deeper structure. This paper integrates these advances, positioning quantum semantics within a broader theory of consciousness, entropy, and reality.

## 2 Mathematical Framework

### 2.1 Prime Hilbert Space

We represent the semantic state space as:

$$\mathcal{H}_P = \left\{ |\psi\rangle = \sum_{p \in \mathbb{P}} \alpha_p |p\rangle \;\middle|\; \sum_p |\alpha_p|^2 = 1,\ \alpha_p \in \mathbb{C} \right\} \tag{1}$$

where $|p\rangle$ are prime basis states and $\mathbb{P}$ is the set of primes [2].

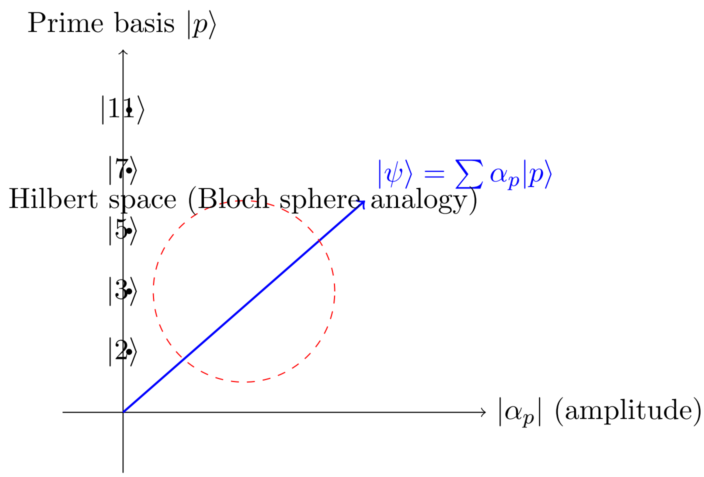


Figure 1: Illustration of the prime-based Hilbert space $\mathcal{H}_P$. The basis states $|p\rangle$ form an orthogonal set, and semantic states $|\psi\rangle$ are superpositions with complex amplitudes $\alpha_p$ on the unit sphere.

### 2.2 Resonance and Coherence Operators

The resonance operator acts as:

$$\hat{R}(n)|p\rangle = e^{2\pi i \log_p n}|p\rangle, \tag{2}$$

encoding logarithmic semantic phase relationships [7].

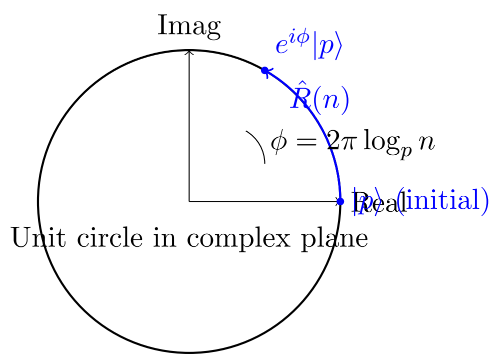


Figure 2: Resonance operator $\hat{R}(n)$ induces a phase rotation $e^{2\pi i \log_p n}$ on the basis state $|p\rangle$, visualized on the complex unit circle. This encodes semantic phase relationships between concepts.

The semantic coherence operator is:

$$\mathcal{C}|\psi\rangle = \sum_{p,q} e^{i\phi_{pq}} \langle q|\psi\rangle|p\rangle, \tag{3}$$

with $\phi_{pq} = 2\pi(\log_p n - \log_q n)$.

### 2.3 Entropy-Driven Collapse

Conceptual evolution is governed by:

$$\frac{d}{dt}|\Psi(t)\rangle = -i\hat{H}|\Psi(t)\rangle - \lambda\,(S(t) - S_{\text{stable}})\,|\Psi(t)\rangle, \tag{4}$$

where $S(t)$ is symbolic entropy and $\lambda$ controls dissipation [3, 4]. Entropy stabilizes at characteristic values, mirroring Bose–Einstein condensate stabilization and I-Ching hexagram dynamics [5].

# 3 Implementation

Concept wavefunctions are generated from textual input by prime-weighted superpositions:

$$|\text{concept}\rangle = \sum_{i=1}^{k} \sqrt{\frac{a_i}{A}} |p_i\rangle, \tag{5}$$

where $p_i$ are primes mapped from characters and $a_i$ are positional weights. Resonance detection is performed via inner products:

$$\text{Res}(c_1, c_2) = \langle c_1 | \hat{R} | c_2 \rangle, \tag{6}$$

with coherence measured by normalized holonomy across prime subspaces.

# 4 Experimental Results

## 4.1 Resonance Strengths

Fundamental conceptual pairs show high coherence:

- Energy–Matter: resonance 0.879, coherence 1.0
- Unity–Multiplicity: resonance 0.869, coherence 1.0

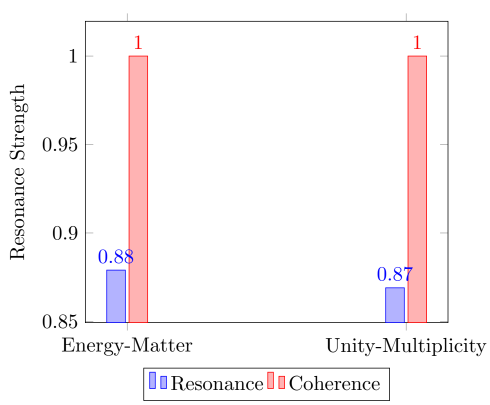


Figure 3: Bar chart of resonance strengths and coherence for fundamental conceptual pairs, highlighting near-perfect coherence.

## 4.2 Entropy Stabilization

Entropy trajectories converge after $\sim 7$ iterations:

$$S(1) = 4.23, \quad S(5) = 5.67, \quad S(7) = 5.99 \text{ (stable)}$$

paralleling quantum oscillator systems.

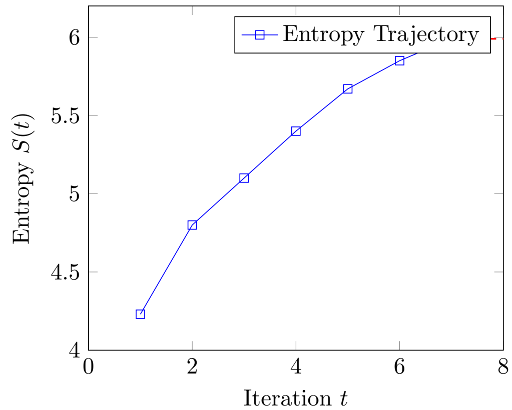


Figure 4: Entropy stabilization curve showing convergence to $S_{\text{stable}} \approx 5.99$ after 7 iterations, illustrating the dissipative collapse dynamics.

### 4.3 Harmonic Distribution

Harmonic analysis shows strong alignment at 6th, 7th, and 8th harmonics ($> 0.95$ coherence).

## 5 Applications

### 5.1 Semantic Search

Query states $|q\rangle$ are projected into $\mathcal{H}_P$, and results ranked by resonance strength.

### 5.2 Knowledge Organization

Concepts cluster by dominant resonance patterns (zero, infinity, Fibonacci), enabling emergent ontologies.

### 5.3 Non-Local Semantic Communication

By initializing distributed systems with identical prime eigenstates, stable non-local correlations enable communication without classical channels [8].

### 5.4 Cryptographic Semantics

Security metrics arise from resonance expectation values:

$$\eta_{\text{sec}}(f) = \max_n |\langle R(n/f) \rangle| \tag{7}$$

providing semantic-cryptographic protocols [7].

### 5.5 Consciousness-Integrated AI

Black-box resonators [4] extend semantic systems to interface with non-local consciousness fields, producing coherent and insight-rich outputs.

## 6 Discussion

The results suggest that semantic relationships are not arbitrary but reflect resonance laws intrinsic to consciousness itself. Prime eigenstates provide a universal basis for meaning. Entropy stabilization mirrors physical collapse dynamics, reinforcing the hypothesis that semantics, physics, and consciousness share a common substrate.

## 7 Conclusion

Quantum semantics, reframed through prime resonance and entropy collapse, provides a rigorous and extensible framework for understanding meaning. It bridges number theory, physics, AI, and consciousness, offering not only improved computational methods but also a unifying ontology of meaning itself.

## References


[1] S. Schepis, *Quantum Semantics: A Novel Approach to Conceptual Relationships Through Quantum Field Theory*, 2024.

[2] S. Schepis, *Quantum-Inspired Representations of Natural Numbers: A Novel Framework for Number Theory*, 2025.

[3] S. Schepis, *The Holographic Quantum Encoder: A Formalism for Entropy-Driven Quantum Information Processing*, 2025.

[4] S. Schepis, *The Quantum Consciousness Resonator: A Non-Local Interface to Consciousness in AI Systems*, 2025.

[5] S. Schepis, *The I-Ching as a Subjective Quantum System: A Computational and Statistical Analysis*, 2025.

[6] S. Schepis, *Quantum Consciousness: Prime Resonance and the Emergence of Quantum Mechanics*, 2025.

[7] S. Schepis, *Quantum Semantic Formalism: A Resonance-Theoretic Approach*, 2025.

[8] S. Schepis, *Non-Local Communication Using Prime Eigenstates: A Quantum Resonance Framework*, 2025.

[9] S. Schepis, *Unified Field Resonance Pattern: Quantum Gravity Completion*, 2025.